

Samalka Anandagoda, Ph.D.

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SUMMARY

- Proven ability to extract, clean and analyze large data sets using statistical methods to deliver valuable insights.
- Strong programming skills (python, C++) used for data manipulation, scientific computing, visualization and building computational models for astrophysics research.
- Excellent track record of communicating relevant research work conducted using these techniques at multiple international conferences and reputed journal publications (6+ peer-reviewed publications, full list at: <https://samalkapiy.github.io/publications/>).

RELEVANT RESEARCH AND WORK EXPERIENCE

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|---------------------|---|------------------------------------|
| Aug 2024 - Present | Visiting Assistant Professor of Physics | Washington College, Maryland |
| | <ul style="list-style-type: none">○ Conducted quantitative analysis on ~20 GB of data using pandas and numpy, applying statistical methods and visualization techniques (matplotlib) to extract trends and patterns to answer key questions in astrophysics research. | |
| Aug 2023 – Aug 2024 | Volunteer Research Fellow | Clemson University, SC |
| | <ul style="list-style-type: none">○ Contributed to the development of a GitHub-hosted computational package for simulating multi-source neutrino emissions across the Milky Way Galaxy (Link) utilizing probability distribution functions (PDFs), python programming, and matplotlib. | |
| May 2022 - Aug 2023 | Research Assistant | Los Alamos National Laboratory, NM |
| | <ul style="list-style-type: none">○ Developed and debugged a Monte Carlo particle transport simulation code written in C++. The code incorporates advanced computational algorithms and physics-based simulations to handle complex particle interactions and dynamics. The code was run on a High Performance Computing (HPC) cluster to leverage parallel processing and to increase simulation throughput. | |
| Aug 2016 - Aug 2023 | Astrophysics Researcher (Ph.D. Candidate) | Clemson University, SC |
| | <ul style="list-style-type: none">○ Building computational models to answer astrophysical research questions. This project utilized programming skills such as python, parameter fitting using maximum likelihood estimations, writing SQL queries to obtain data from astrophysical catalogs, data visualization and model building. | |

SOFTWARE RELEASES

SNUGGY: Simulation package that maps the location of the stars in the Milky Way galaxy based on a 2D probability distribution function (PDF) to understand particle interaction processes in the Galaxy. The neutrino and gamma-ray emission are also modeled to ensure the simulated sources follow a log-normal distribution along with a capacity to add more functionality.

SKILLS

- | | |
|----------|---|
| Tools | Python (PyTorch, SciKit, pandas, matplotlib, numpy, scipy, seaborn), C++, SQL, Jupyter Notebooks, Google Colab. |
| Concepts | Statistics, Machine Learning, Clustering algorithms, Regression and Classification, Differential equations, Monte Carlo method. |
| | Git version control, High performance computing. |

LICENSES AND CERTIFICATIONS

IBM Data Science Professional Certificate

- Issued : July 1st 2025, Credential ID: [A7TEPLV81V4K](#)
Skills : hands-on experience in data preprocessing, exploratory data analysis (EDA), advanced visualization techniques, and the development of end-to-end machine learning pipelines. Data handling using SQL, pandas. Develop and optimize Decision Tree classifiers; implement K-Nearest Neighbors (KNN) for supervised classification tasks. Train and evaluate Random Forest and XGBoost regression models for performance comparison using cross-validation and metrics such as RMSE, R^2 , and MAE.

EDUCATION

Aug 2019 - Aug 2023	Ph.D. in Astronomy and Astrophysics	Clemson University, SC
Aug 2016 - Aug 2019	MS in Astronomy and Astrophysics	Clemson University, SC
Nov 2011 - Nov 2015	Bachelor of Science in Physics Major: Physics, Minor: Computer Science and Mathematics	University of Peradeniya, Sri Lanka

EXTRACURRICULAR AND COCURRICULAR ACTIVITIES

- Served as a referee for multiple papers at the Astrophysical Journal (APJ).
- Co-organizer for the 2019 MASC (Meeting for astronomers in South Carolina) meeting held at Clemson University.